

ITBAYAT, Philippines

WMO: 981320

Lat: 20.787N

Lon: 121.840E

Elev: 407

StdP: 14.48

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	63.9	65.2	55.8	67.4	67.0	58.0	73.0	67.1	19.7	67.1	18.4	69.5	12.5	40	0.361

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	7.4	89.1	83.3	88.1	82.9	87.4	82.5	84.2	88.1	83.6	87.4	82.9	86.7	5.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.0	174.7	87.5	82.4	171.6	87.0	81.8	168.0	86.3	48.7	88.4	48.0	87.7	47.3	87.0	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 21.0	17.3	15.0	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	57.8	85.4	4.1	1.0	54.9	86.1	52.5	86.6	50.2	87.1	47.2	87.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.1	71.5	72.5	74.3	78.1	81.1	83.2	82.9	82.2	81.3	79.4	77.0	73.0
	DBStd	4.93	3.09	3.42	3.69	3.27	2.56	2.15	1.94	1.94	1.71	2.06	2.47	3.31
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	1	1	0	0	0	0	0	0	0	0	0	0
	CDD50	10242	666	630	752	843	965	995	1020	997	939	913	809	713
	CDD65	4770	201	212	288	393	500	545	555	532	489	448	359	248
	CDH74	41358	302	517	1215	3032	5162	6464	6580	6046	5214	3931	2202	693
	CDH80	10540	6	18	109	534	1422	2363	2317	1838	1277	526	120	10
(14) Wind	WSAvg	7.9	9.6	8.7	8.3	7.3	5.7	5.9	6.9	6.9	7.2	9.0	9.1	10.1
(15) Precipitation	PrecAvg	95.6	6.1	3.4	2.3	2.2	8.4	8.1	10.2	13.3	13.1	11.7	9.9	6.0
	PrecMax	132.8	14.5	10.6	5.1	8.9	24.3	19.2	19.5	23.9	35.9	24.8	31.8	15.5
	PrecMin	61.8	1.1	0.7	0.5	0.2	0.3	0.4	1.3	3.1	1.9	0.9	1.9	1.3
	PrecStd	17.0	3.6	2.4	1.2	1.8	5.6	4.8	5.5	5.9	7.3	6.4	6.8	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.3	82.0	85.3	87.4	89.6	90.7	89.8	88.9	87.6	86.1	84.0	81.6
		MCWB	74.8	77.1	78.8	81.0	83.3	84.2	83.8	83.5	82.6	81.2	79.8	75.3
	2%	DB	78.1	80.0	82.7	85.9	88.1	89.2	88.7	87.8	86.4	84.6	82.6	79.7
		MCWB	73.9	75.6	77.7	80.5	82.8	83.6	83.2	83.0	81.8	80.3	78.8	75.9
	5%	DB	76.8	78.5	80.8	84.5	86.9	88.2	87.7	86.9	85.5	83.7	81.5	78.4
		MCWB	73.1	74.6	76.7	79.6	82.2	83.0	82.6	82.5	81.2	79.7	78.0	75.1
	10%	DB	75.6	77.0	79.2	83.1	85.8	87.2	86.8	86.0	84.6	82.7	80.5	77.3
		MCWB	72.1	73.5	75.5	78.7	81.4	82.5	82.1	81.9	80.5	78.9	77.2	74.2
	0.4%	WB	76.1	77.9	79.7	82.2	84.3	85.2	84.6	84.4	83.6	81.9	80.6	78.1
		MCDB	79.1	81.1	84.0	85.9	88.2	89.4	88.5	87.6	86.6	84.9	83.3	79.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	74.7	76.4	78.2	80.9	83.4	84.3	83.8	83.6	82.5	81.0	79.3	76.8
		MCDB	77.4	79.4	81.9	84.9	87.3	88.4	87.9	87.0	85.7	83.8	81.9	78.8
	5%	WB	73.6	75.1	77.0	80.1	82.6	83.6	83.2	82.8	81.7	80.3	78.5	75.7
		MCDB	76.1	78.0	80.3	83.9	86.5	87.6	87.1	86.2	84.9	83.1	81.0	77.8
	10%	WB	72.7	73.8	75.9	79.0	81.7	82.8	82.5	82.1	81.0	79.5	77.7	74.8
		MCDB	75.2	76.5	78.9	82.5	85.4	86.7	86.3	85.5	84.1	82.2	80.1	77.0
	5% DB	MDBR	5.3	6.2	6.5	7.2	7.5	7.4	7.3	6.8	6.3	5.3	4.7	4.8
		MCDBR	6.8	8.4	8.7	9.0	9.5	8.6	8.4	8.1	7.4	6.2	5.9	6.3
		MCWBR	5.1	6.1	5.4	5.3	5.9	5.1	5.2	5.1	4.9	4.4	4.2	4.4
		MCWBR	6.0	7.8	8.1	8.5	9.1	8.2	7.8	7.6	7.0	6.1	5.6	5.3
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.2	6.2	5.6	5.3	6.2	5.4	5.3	5.1	5.0	4.7	4.3	4.3
(40) Clear-Sky Solar Irradiance	taub		0.451	0.453	0.537	0.523	0.455	0.449	0.450	0.470	0.483	0.482	0.480	0.487
	taud		2.243	2.267	2.023	2.113	2.392	2.422	2.411	2.338	2.271	2.227	2.215	2.132
	Ebn at Noon		259	267	248	253	269	268	268	264	259	255	248	242
	Edh at Noon		41	42	56	52	39	37	38	41	43	43	42	44
(44) All-Sky Solar Radiation	RadAvg		1043	1335	1530	1834	1881	1971	1904	1717	1623	1468	1148	928
	RadStd		107	175	154	182	157	169	161	157	172	91	121	140

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page